

## Employee

The screenshot shows the DBeaver 26.1.5 interface. The SQL Editor contains the following script:

```
85 CREATE TABLE DEPENDENT (  
86     Essn CHAR(9) NOT NULL,  
87     Dependent_name VARCHAR(20) NOT NULL,  
88     Sex CHAR(1) CHECK (Sex IN ('M', 'F')),  
89     Bdate DATE,  
90     Relationship VARCHAR(15),  
91     CONSTRAINT pk_dependent PRIMARY KEY (Essn, Dependent_name),  
92     CONSTRAINT fk_dependent_emp FOREIGN KEY (Essn)  
93         REFERENCES EMPLOYEE (Ssn)  
94         ON DELETE CASCADE  
95         ON UPDATE CASCADE  
96 );  
97  
98 DESC EMPLOYEE;  
99 -- DESC DEPARTMENT;  
100 -- DESC DEPT_LOCATIONS;  
101 -- DESC PROJECT;  
102 -- DESC WORKS_ON;  
103 -- DESC DEPENDENT;
```

Below the script, the 'columns 1' tab is active, displaying a table grid with the following data:

| AZ Field    | AZ Type       | AZ Null | AZ Key | AZ Default | AZ Extra |
|-------------|---------------|---------|--------|------------|----------|
| 1 Fname     | varchar(15)   | NO      |        | [NULL]     |          |
| 2 Minit     | char(1)       | YES     |        | [NULL]     |          |
| 3 Lname     | varchar(15)   | NO      |        | [NULL]     |          |
| 4 Ssn       | char(9)       | NO      | PRI    | [NULL]     |          |
| 5 Bdate     | date          | YES     |        | [NULL]     |          |
| 6 Address   | varchar(50)   | YES     |        | [NULL]     |          |
| 7 Sex       | char(1)       | YES     |        | [NULL]     |          |
| 8 Salary    | decimal(10,2) | YES     |        | [NULL]     |          |
| 9 Super_ssn | char(9)       | YES     | MUL    | [NULL]     |          |
| 10 Dno      | int           | YES     | MUL    | [NULL]     |          |

The bottom status bar indicates '10 row(s) fetched - 2.4' and '98...81'.

## Department

The screenshot shows the DBeaver 26.1.5 interface. The main editor displays an SQL script for creating a table named `DEPENDENT`. The script includes a primary key constraint `pk_dependent` on `Essn` and `Dependent_name`, and a foreign key constraint `fk_dependent_emp` on `Essn` referencing the `EMPLOYEE` table. Below the script, the 'columns 1' tab is active, showing the table structure for `DESC DEPARTMENT`.

|   | AZ Field       | AZ Type     | AZ Null | AZ Key | AZ Default | AZ Extra |
|---|----------------|-------------|---------|--------|------------|----------|
| 1 | Dname          | varchar(25) | NO      | UNI    | [NULL]     |          |
| 2 | Dnumber        | int         | NO      | PRI    | [NULL]     |          |
| 3 | Mgr_ssn        | char(9)     | NO      | MUL    | [NULL]     |          |
| 4 | Mgr_start_date | date        | YES     |        | [NULL]     |          |

The bottom status bar indicates '4 row(s) fetched - 2.31' and '99...99'.

department locations

The screenshot shows the DBeaver 26.1.5 interface. The main window displays a SQL script for creating a table named `DEPENDENT` and describing several tables. The script is as follows:

```
85 CREATE TABLE DEPENDENT (  
86     Essn CHAR(9) NOT NULL,  
87     Dependent_name VARCHAR(20) NOT NULL,  
88     Sex CHAR(1) CHECK (Sex IN ('M', 'F')),  
89     Bdate DATE,  
90     Relationship VARCHAR(15),  
91     CONSTRAINT pk_dependent PRIMARY KEY (Essn, Dependent_name),  
92     CONSTRAINT fk_dependent_emp FOREIGN KEY (Essn)  
93         REFERENCES EMPLOYEE (Ssn)  
94         ON DELETE CASCADE  
95         ON UPDATE CASCADE  
96 );  
97  
98 -- DESC EMPLOYEE;  
99 -- DESC DEPARTMENT;  
100 DESC DEPT_LOCATIONS;  
101 -- DESC PROJECT;  
102 -- DESC WORKS_ON;  
103 -- DESC DEPENDENT;
```

The output pane on the right shows a list of tables, all marked as "Unknown table".

Below the script, the "columns 1" tab is active, showing the description of the `DEPT_LOCATIONS` table. The table has two columns:

| AZ Field  | AZ Type     | AZ Null | AZ Key | AZ Default | AZ Extra |
|-----------|-------------|---------|--------|------------|----------|
| Dnumber   | int         | NO      | PRI    | [NULL]     |          |
| Dlocation | varchar(20) | NO      | PRI    | [NULL]     |          |

The bottom status bar indicates "2 row(s) fetched - 2.37".

## Project

The screenshot shows the DBeaver 26.1.5 interface. The main window displays a SQL script for creating a table named `DEPENDENT`. The script includes column definitions for `Essn`, `Dependent_name`, `Sex`, `Bdate`, and `Relationship`, along with primary and foreign key constraints. Below the script, the `DESC PROJECT` command is executed, resulting in a table structure view.

**SQL Script:**

```
85 CREATE TABLE DEPENDENT (  
86     Essn CHAR(9) NOT NULL,  
87     Dependent_name VARCHAR(20) NOT NULL,  
88     Sex CHAR(1) CHECK (Sex IN ('M', 'F')),  
89     Bdate DATE,  
90     Relationship VARCHAR(15),  
91     CONSTRAINT pk_dependent PRIMARY KEY (Essn, Dependent_name),  
92     CONSTRAINT fk_dependent_emp FOREIGN KEY (Essn)  
93         REFERENCES EMPLOYEE (Ssn)  
94         ON DELETE CASCADE  
95         ON UPDATE CASCADE  
96 );  
97  
98 -- DESC EMPLOYEE;  
99 -- DESC DEPARTMENT;  
100 -- DESC DEPT_LOCATIONS;  
101 DESC PROJECT;  
102 -- DESC WORKS_ON;  
103 -- DESC DEPENDENT;
```

**Table Structure (DESC PROJECT):**

| AZ Field    | AZ Type     | AZ Null | AZ Key | AZ Default | AZ Extra |
|-------------|-------------|---------|--------|------------|----------|
| 1 Pname     | varchar(25) | NO      | UNI    | [NULL]     |          |
| 2 Pnumber   | int         | NO      | PRI    | [NULL]     |          |
| 3 Plocation | varchar(20) | YES     |        | [NULL]     |          |
| 4 Dnum      | int         | NO      | MUL    | [NULL]     |          |

The interface also shows a sidebar with navigation options (Files, Bookmarks, Dashboards, Diagrams, Scripts) and a bottom status bar indicating the current table is `PROJECT` with 4 rows fetched.

## Works On

The screenshot shows the DBeaver 26.1.5 interface. The main window displays a SQL script for creating a table named `DEPENDENT`. The script includes a primary key constraint on `Essn` and `Dependent_name`, and a foreign key constraint on `Essn` referencing the `EMPLOYEE` table. Below the script, the `DESC WORKS_ON` command is executed, and the results are displayed in a table.

**SQL Script:**

```
85 CREATE TABLE DEPENDENT (  
86     Essn CHAR(9) NOT NULL,  
87     Dependent_name VARCHAR(20) NOT NULL,  
88     Sex CHAR(1) CHECK (Sex IN ('M', 'F')),  
89     Bdate DATE,  
90     Relationship VARCHAR(15),  
91     CONSTRAINT pk_dependent PRIMARY KEY (Essn, Dependent_name),  
92     CONSTRAINT fk_dependent_emp FOREIGN KEY (Essn)  
93         REFERENCES EMPLOYEE (Ssn)  
94         ON DELETE CASCADE  
95         ON UPDATE CASCADE  
96 );  
97  
98 -- DESC EMPLOYEE;  
99 -- DESC DEPARTMENT;  
100 -- DESC DEPT_LOCATIONS;  
101 -- DESC PROJECT;  
102 DESC WORKS_ON;  
103 -- DESC DEPENDENT;
```

**Table Structure (DESC WORKS\_ON):**

| AZ Field | AZ Type      | AZ Null | AZ Key | AZ Default | AZ Extra |
|----------|--------------|---------|--------|------------|----------|
| 1 Essn   | char(9)      | NO      | PRI    | [NULL]     |          |
| 2 Pno    | int          | NO      | PRI    | [NULL]     |          |
| 3 Hours  | decimal(4,1) | YES     |        | [NULL]     |          |

The interface also shows a sidebar with navigation options (General, Bookmarks, Dashboards, Diagrams, Scripts) and a bottom status bar indicating the current table is `EDT en Writable` and `Smart Insert` is enabled.

## Dependent

The screenshot displays the DBeaver 26.1.5 SQL Editor. The main window shows the SQL script for creating a table named 'DEPENDENT'. The script includes the following SQL code:

```
CREATE TABLE DEPENDENT (  
  Essn CHAR(9) NOT NULL,  
  Dependent_name VARCHAR(20) NOT NULL,  
  Sex CHAR(1) CHECK (Sex IN ('M', 'F')),  
  Bdate DATE,  
  Relationship VARCHAR(15),  
  CONSTRAINT pk_dependent PRIMARY KEY (Essn, Dependent_name),  
  CONSTRAINT fk_dependent_emp FOREIGN KEY (Essn)  
    REFERENCES EMPLOYEE (Ssn)  
    ON DELETE CASCADE  
    ON UPDATE CASCADE  
);  
  
-- DESC EMPLOYEE;  
-- DESC DEPARTMENT;  
-- DESC DEPT_LOCATIONS;  
-- DESC PROJECT;  
-- DESC WORKS_ON;  
DESC DEPENDENT;
```

Below the script, the 'DESC DEPENDENT' command is executed, and the results are displayed in a grid view. The grid shows the table structure with columns: AZ Field, AZ Type, AZ Null, AZ Key, AZ Default, and AZ Extra. The data is as follows:

| AZ Field         | AZ Type     | AZ Null | AZ Key | AZ Default | AZ Extra |
|------------------|-------------|---------|--------|------------|----------|
| 1 Essn           | char(9)     | NO      | PRI    | [NULL]     |          |
| 2 Dependent_name | varchar(20) | NO      | PRI    | [NULL]     |          |
| 3 Sex            | char(1)     | YES     |        | [NULL]     |          |
| 4 Bdate          | date        | YES     |        | [NULL]     |          |
| 5 Relationship   | varchar(15) | YES     |        | [NULL]     |          |

The bottom status bar indicates '5 row(s) fetched - 2.24' and '10...78'.

IDK if this is physically possible in the course of an hour, prof, unless you know exactly what you are doing.